

EXERCISE SET 2.1*

Write the statements in 6-9 in symbolic form using the symbols $\sim$, $\vee$, and $\wedge$ and the indicated letters to represent component statements.

8. Let h = "John is healthy," w = "John is wealthy," and s = "John is wise."
- a. John is healthy and wealthy but not wise.
 - b. John is not wealthy but he is healthy and wise.
 - c. John is neither healthy, wealthy, nor wise.
 - d. John is neither wealthy nor wise, but he is healthy.
 - e. John is wealthy, but he is not both healthy and wise.

Solution: a. $(h \wedge w) \wedge \sim s$

 b. $\sim w \wedge (h \wedge s)$

 c. $\sim(h \vee w \vee s) \equiv \sim h \wedge \sim w \wedge \sim s$

 d. $\sim(w \vee s) \wedge h \equiv \sim w \wedge \sim s \wedge h$

 e. $w \wedge \sim(h \wedge s) \equiv w \wedge (\sim h \vee \sim s)$